

Short Presentation

Apples and pears? Impacts of using the Jacobian scaled conditional Akaike Information Criteria for model selection in linear mixed models. Advanced Small Area Estimation (Prof. Schmid)

Niklas Ippisch

Background & Motivation

- ▶ Choosing an adequate set of variables essential for good model fit
- ▶ Data driven transformation can improve the model fit
- ▶ Transformations especially important if normality assumption(s) not fulfilled
- ▶ Problem: Variable selection and estimation of transformation parameter λ affect each other:
 - ▶ Optimal set of covariates depends on transformation chosen and vice versa
 - ▶ Other papers suggest the usage of a conditional Akaike Information Criteria: $cAIC = -2\log(g(y|\hat{\theta}, \hat{u})) + 2K_{MLE}$
 - ▶ Problem: It is not scale invariate, therefore cannot be used to compare models with different target variable y

Proposed Solution I

- ▶ Solution: Scaling the $cAIC$ with the Jacobian J :
$$JcAIC = -2\log(g(\tilde{y}|\hat{\theta}, \hat{u})) - 2\log(J(\hat{\lambda}, y)) + 2BC$$
- ▶ $\tilde{y}$ being the vector of transformed y with transformation $T_{\lambda}(y)$
- ▶ g being the density function of the approximating model
- ▶ J being the Jacobian matrix:
$$J(\hat{\lambda}, y) = \prod_{i=1}^D \prod_{j=1}^{N_i} \frac{\partial \tilde{y}_{ij}}{\partial y_{ij}} = \prod_{i=1}^D \prod_{j=1}^{N_i} (y_{ij} + s)^{\lambda-1}$$
- ▶ BC being a bias correction term, estimated by a bootstrap

Proposed Solution II

1. Start with full model with all p variables and estimate $\hat{\lambda}$
2. For each step $s = 1, \dots, S$:
 - i) Consider all possible models which exclude one predictor
 - ii) Estimate $\hat{\lambda}$ for each candidate and calculate $JcAIC$ with estimated $\hat{\lambda}$
 - iii) Compare $JcAIC$ values and choose model with lowest value as new optimal model
3. Compare $JcAIC$ from 2.iii) with the previously optimal model from step $s - 1$. Repeat until no better $JcAIC$ can be found.

Methodical approach for the term paper

- ▶ Research question: Does the proposed simultaneous selection and estimation outperform the naive selection in different data scenarios?
- ▶ Approach I: Simulation Study
 - ▶ Three different distributions (e.g., normal, log, pareto)
 - ▶ Inclusion of two irrelevant and one weak associated variables in the model
 - ▶ Comparison of naive approach vs. proposed solution based on $JcAIC$ and proportion selecting the correct model
- ▶ Approach II: Application to real world data
 - ▶ Data from Mozambique
 - ▶ Estimation of head count ration and poverty gap based on naive approach vs. proposed solution

Pseudo Code for model based simulation

1. Select data settings (Normal, Log, Box-Cox, Pareto?) and data generating model
2. For $m = 1, \dots, M$ iterations:
 - i) Draw covariates from respective distributions and create pseudo population, including irrelevant variable(s)
 - ii) For $a = 1, \dots, A$ variable selection approaches:
 - I) Perform stepwise variable selection and/or estimation of λ
 - II) Calculate $JcAIC$ for optimal model
3. Compare estimated lambda and $JcAIC$ of each optimal model

Discussion

- ▶ Currently still subject to discussion
 - ▶ What distributions to use
 - ▶ Domain and sample sizes (+ association)
 - ▶ Evaluation of real world data application
- ▶ Open questions
 - ▶ Citation of methodology (especially pseudo code)
 - ▶ Comparability with results of Krennmair et al. (2026)

Thank you for your attention!

Literature

Krennmair, P., Würz, N., Schmid, T., & Tzavidis, N. (2026). Random forests and mixed effects random forests for small area estimation of general parameters: A poverty mapping case study in mozambique. *Annals of Applied Statistics*, 20(1), 809–832. <https://doi.org/10.1214/25-AOAS2126>

Lee, Y., Rojas-Perilla, N., Runge, M., & Schmid, T. (2023). Variable selection using conditional AIC for linear mixed models with data-driven transformations. *Statistics and Computing*, 33(1), 27. <https://doi.org/10.1007/s11222-022-10198-9>

Notes: data situation simulation study imitating Mozambique data